



spare-it

# Spare-it: Contamination Identification

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**01**

# **Project Overview**

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# Project Overview



## The Client

**Spare-it** is a startup that aims to empower businesses, office owners, and universities with the ability to monitor and reduce various types of workspace waste effectively.



## Problem Statement

There is a **need for a reliable source of real-time waste consumption data** to allow communities and organizations to reduce their waste consumption and be aware of their waste practices.



## The Goal

Build a solution that allows organizations to continuously monitor their **(1) waste composition, (2) waste contamination, and (3) missed recycling/composting opportunities**.



# Use Cases

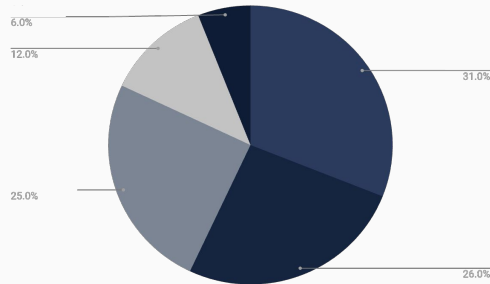


# Use Cases

1

## Waste Composition

Organizations can understand the composition of their waste at a granular level.



2

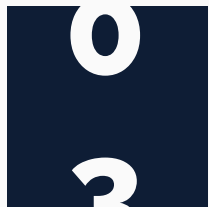
## Waste Contamination

Organizations can understand their rate of waste contamination. For example, a certain floor in an office building may tend to mistakenly compost magazines or recycle black plastics.

3

## Missed Opportunities

Organizations can understand their rate of missed opportunities for recycling and composting and identify common items for which missed opportunities occur in order to raise awareness.



# **Solution Approach**

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# Solution Approach

**Proposed ML Solution:** Train a YOLOv8 segmentation model for waste detection and classification with a focus on achieving high quality training data

## What we implemented:

1. External dataset integration to diversify the objects in the dataset
  2. Copy-paste augmentation to augment underrepresented classes and improve the class imbalance in the dataset
  3. Trained YOLOv8 model for real-time waste segmentation
  4. AI image generation to support synthetic data generation for future improvements
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# Solution Approach: External Dataset Integration (TACO)

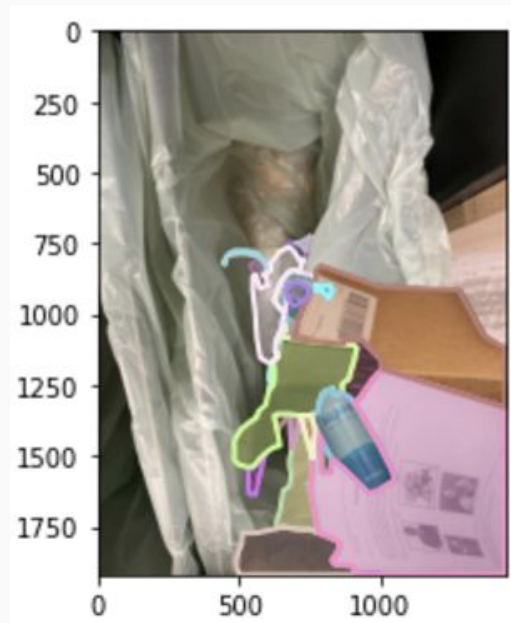


- 1000+ images
- 4700+ segmentations
- Used for copy-paste augmentation

```
31: 24, # Tissues -> Paper Towel/Napkins/Tissue
12: 63, # Drink can -> Metal Can
36: 106, # Plastic film -> Plastic Wrap
34: 83, # Paper bag -> Empty Paper Bag
25: 52, # Food waste -> Other Food or Mixed Food
20: 1, # Paper cup -> Paper Cup
1: 38, # Battery -> Batteries
55: 96, # Plastic straw -> Straws
21: 101, # Disposable plastic cup -> Clear Plastic Cup
30: 80, # Magazine paper -> Magazines Newspaper
49: 98, # Plastic utensils -> Plastic Cutlery
0: 65, # Aluminium foil -> Aluminum Foil
6: 58, # Glass bottle -> Glass Bottles
8: 69, # Metal bottle cap -> Metallic Bottle Cap or Lid
32: 118 # Wrapping paper -> Wrapping Paper
```

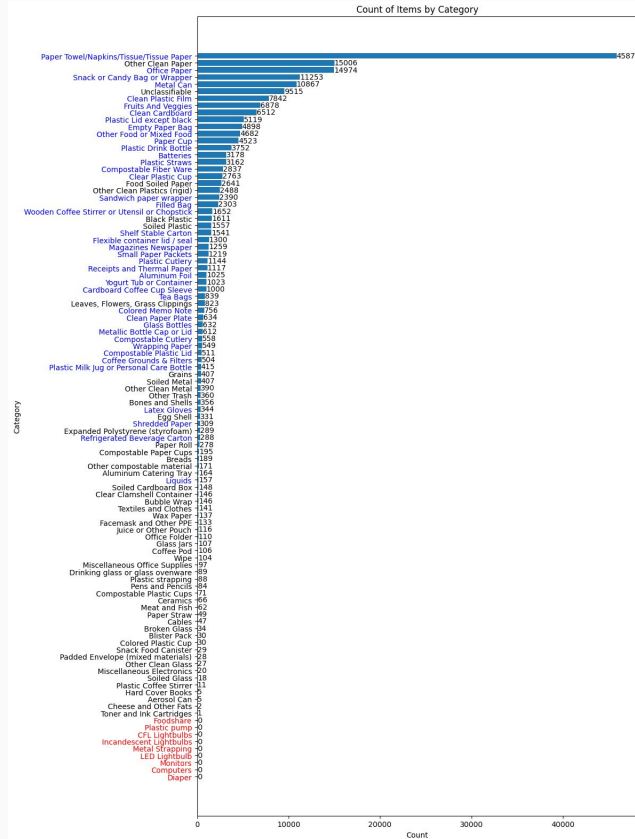
# Solution Approach: Copy-paste

- Multiple Pasting - Random amount of segments applied randomly to the same base image
- Taco Implementation - Taco dataset images used after they are converted to COCO format
- Class Imbalance - Fine tuning of classes performed to generate a more balanced dataset

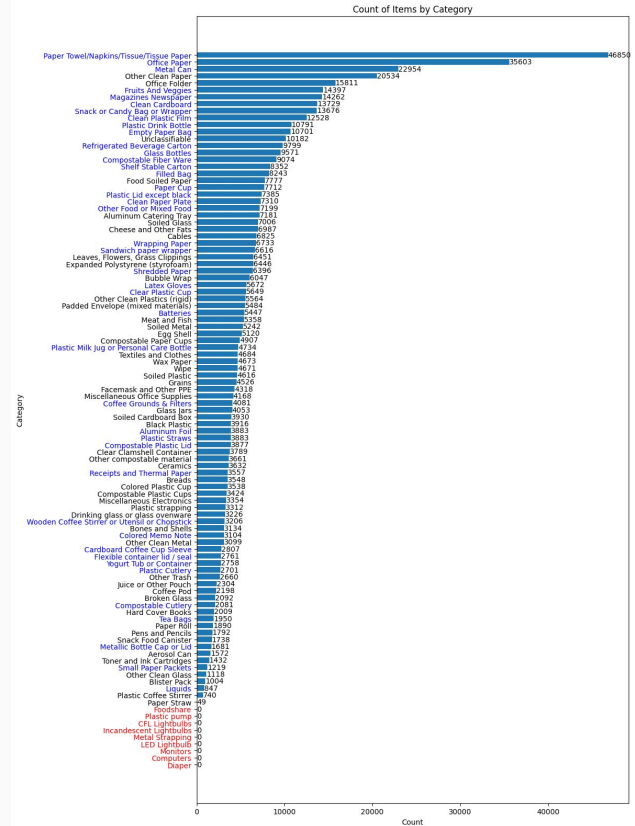


# Solution Approach: Copy-paste

Original  
Dataset



With  
Copy-paste

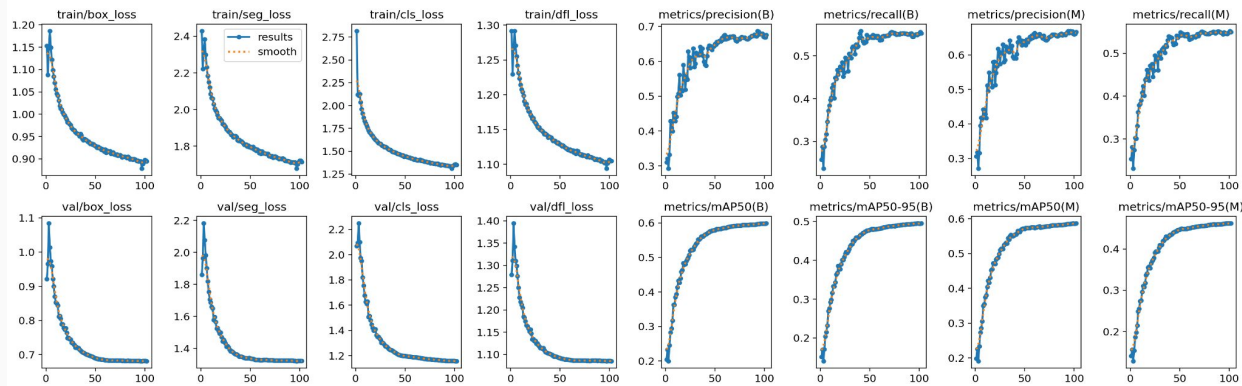


# Solution Approach: YOLOv8 Segmentation Model

**Training Pipeline.** Integrated improved copy-paste augmentation and external datasets and experimented with a variety of training configurations based on the tools we built

## Best Results.

The copy-paste augmentation along with the original Spare-it dataset achieved 0.587 mask mAP@50 compared to baseline 0.539 mask mAP@50



YOLOv8 Training Results

# Solution Approach: YOLOv8 Segmentation

Baseline Model Performance (Original Spare-it Dataset)

| Class           | Images       | Instances   | Box (P)    | Box (R)       | Box (mAP50)   | Box (mAP50-95) | Mask (P)    | Mask (R)       | Mask (mAP50) | Mask (mAP50-95) |
|-----------------|--------------|-------------|------------|---------------|---------------|----------------|-------------|----------------|--------------|-----------------|
| Paper Towel     | 906          | 1870        | 0.67       | 0.671         | 0.705         | 0.534          | 0.671       | 0.673          | 0.704        | 0.501           |
| Office Paper    | 248          | 562         | 0.481      | 0.626         | 0.564         | 0.467          | 0.458       | 0.6            | 0.525        | 0.405           |
| Snack or Can    | 501          | 758         | 0.679      | 0.534         | 0.623         | 0.478          | 0.673       | 0.53           | 0.612        | 0.452           |
| Metal Can       | 343          | 634         | 0.821      | 0.886         | 0.912         | 0.806          | 0.824       | 0.89           | 0.916        | 0.813           |
| Clean Plastic   | 427          | 626         | 0.572      | 0.5           | 0.507         | 0.347          | 0.565       | 0.495          | 0.505        | 0.316           |
| Fruits And V    | 127          | 219         | 0.395      | 0.557         | 0.47          | 0.343          | 0.394       | 0.558          | 0.471        | 0.312           |
| Clean Cardb     | 364          | 573         | 0.576      | 0.63          | 0.622         | 0.509          | 0.569       | 0.623          | 0.612        | 0.493           |
| Plastic Lid ex  | 306          | 372         | 0.619      | 0.558         | 0.573         | 0.49           | 0.616       | 0.556          | 0.561        | 0.439           |
| Empty Paper     | 460          | 569         | 0.607      | 0.663         | 0.65          | 0.534          | 0.601       | 0.657          | 0.639        | 0.516           |
| Other Food c    | 133          | 196         | 0.54       | 0.396         | 0.377         | 0.252          | 0.544       | 0.403          | 0.384        | 0.216           |
| Paper Cup       | 215          | 259         | 0.685      | 0.699         | 0.73          | 0.653          | 0.7         | 0.714          | 0.742        | 0.64            |
| Plastic Drink   | 265          | 355         | 0.795      | 0.764         | 0.82          | 0.698          | 0.792       | 0.761          | 0.819        | 0.695           |
| Batteries       | 1            | 1           | 1          | 0             | 0             | 0              | 1           | 0              | 0            | 0               |
| Plastic Straw   | 148          | 164         | 0.734      | 0.61          | 0.68          | 0.52           | 0.732       | 0.61           | 0.677        | 0.425           |
| Compostable     | 146          | 196         | 0.654      | 0.703         | 0.699         | 0.623          | 0.649       | 0.699          | 0.703        | 0.566           |
| Clear Plastic   | 167          | 189         | 0.735      | 0.748         | 0.772         | 0.689          | 0.735       | 0.749          | 0.77         | 0.692           |
| Sandwich pa     | 126          | 170         | 0.696      | 0.559         | 0.627         | 0.474          | 0.701       | 0.565          | 0.637        | 0.453           |
| Filled Bag      | 203          | 223         | 0.584      | 0.471         | 0.519         | 0.42           | 0.581       | 0.471          | 0.507        | 0.38            |
| Wooden Coff     | 53           | 65          | 0.611      | 0.628         | 0.624         | 0.501          | 0.597       | 0.614          | 0.598        | 0.376           |
| Shelf Stable    | 62           | 135         | 0.725      | 0.696         | 0.73          | 0.621          | 0.724       | 0.696          | 0.725        | 0.607           |
| Flexible cont   | 74           | 91          | 0.698      | 0.297         | 0.38          | 0.32           | 0.671       | 0.286          | 0.367        | 0.298           |
| Magazines N     | 76           | 112         | 0.559      | 0.656         | 0.614         | 0.477          | 0.544       | 0.64           | 0.586        | 0.388           |
| Small Paper     | 22           | 25          | 0.171      | 0.12          | 0.043         | 0.035          | 0.171       | 0.12           | 0.042        | 0.033           |
| Plastic Cutlery | 65           | 78          | 0.389      | 0.231         | 0.239         | 0.186          | 0.409       | 0.244          | 0.23         | 0.153           |
| Receipts and    | 65           | 71          | 0.358      | 0.282         | 0.224         | 0.199          | 0.376       | 0.296          | 0.238        | 0.199           |
| Aluminum Fo     | 45           | 51          | 0.653      | 0.701         | 0.746         | 0.636          | 0.653       | 0.702          | 0.749        | 0.604           |
| Yogurt Tub o    | 47           | 57          | 0.702      | 0.491         | 0.626         | 0.537          | 0.701       | 0.491          | 0.585        | 0.461           |
| Cardboard C     | 65           | 67          | 0.784      | 0.716         | 0.739         | 0.615          | 0.784       | 0.716          | 0.739        | 0.62            |
| Tea Bags        | 32           | 33          | 0.52       | 0.121         | 0.168         | 0.099          | 0.518       | 0.121          | 0.197        | 0.121           |
| Colored Men     | 44           | 54          | 0.502      | 0.187         | 0.234         | 0.203          | 0.503       | 0.188          | 0.233        | 0.197           |
| Clean Paper     | 41           | 56          | 0.618      | 0.393         | 0.505         | 0.435          | 0.598       | 0.375          | 0.469        | 0.395           |
| Glass Bottles   | 70           | 117         | 0.653      | 0.607         | 0.652         | 0.476          | 0.643       | 0.598          | 0.646        | 0.483           |
| Metallic Bottl  | 49           | 59          | 0.51       | 0.407         | 0.419         | 0.301          | 0.51        | 0.407          | 0.397        | 0.251           |
| Compostable     | 34           | 37          | 0.597      | 0.486         | 0.555         | 0.515          | 0.596       | 0.486          | 0.55         | 0.484           |
| Wrapping Pa     | 95           | 114         | 0.607      | 0.298         | 0.43          | 0.269          | 0.552       | 0.272          | 0.381        | 0.218           |
| Compostable     | 37           | 39          | 0.618      | 0.513         | 0.575         | 0.47           | 0.617       | 0.513          | 0.556        | 0.427           |
| Coffee Groun    | 71           | 90          | 0.86       | 0.753         | 0.863         | 0.736          | 0.861       | 0.754          | 0.868        | 0.729           |
| Plastic Milk J  | 32           | 33          | 0.735      | 0.505         | 0.64          | 0.577          | 0.736       | 0.506          | 0.64         | 0.585           |
| Latex Gloves    | 19           | 29          | 0.59       | 0.596         | 0.707         | 0.536          | 0.592       | 0.6            | 0.704        | 0.559           |
| Shredded Pa     | 63           | 78          | 0.824      | 0.782         | 0.809         | 0.713          | 0.824       | 0.782          | 0.813        | 0.729           |
| Refrigerated    | 17           | 21          | 0.801      | 0.524         | 0.585         | 0.442          | 0.801       | 0.524          | 0.605        | 0.458           |
| Liquids         | 8            | 8           | 0.927      | 0.375         | 0.446         | 0.248          | 0.309       | 0.125          | 0.191        | 0.073           |
| Overall:        | Box(Precisio | Box(Recall) | Box(mAP50) | Box(mAP50-95) | Mask(Precisio | Mask(Recall)   | Mask(mAP50) | Mask(mAP50-95) |              |                 |
|                 | 0.632        | 0.516       | 0.551      | 0.446         | 0.614         | 0.508          | 0.539       | 0.417          |              |                 |

Best Model Performance

| Class           | Images       | Instances   | Box(P)     | Box(R)        | Box(mAP50)    | Box(mAP50-95) | Mask(P)     | Mask(R)        | Mask(mAP50) | Mask(mAP50-95) |
|-----------------|--------------|-------------|------------|---------------|---------------|---------------|-------------|----------------|-------------|----------------|
| Paper Towel     | 906          | 1870        | 0.695      | 0.665         | 0.726         | 0.56          | 0.692       | 0.668          | 0.72        | 0.527          |
| Office Paper    | 248          | 562         | 0.506      | 0.63          | 0.605         | 0.499         | 0.482       | 0.609          | 0.565       | 0.433          |
| Snack or Can    | 501          | 758         | 0.681      | 0.528         | 0.615         | 0.48          | 0.677       | 0.531          | 0.607       | 0.463          |
| Metal Can       | 343          | 634         | 0.86       | 0.883         | 0.922         | 0.823         | 0.858       | 0.885          | 0.923       | 0.826          |
| Clean Plastic   | 427          | 626         | 0.643      | 0.482         | 0.542         | 0.385         | 0.632       | 0.478          | 0.537       | 0.354          |
| Fruits And V    | 127          | 219         | 0.444      | 0.566         | 0.525         | 0.405         | 0.443       | 0.571          | 0.527       | 0.369          |
| Clean Cardb     | 364          | 573         | 0.611      | 0.626         | 0.64          | 0.524         | 0.606       | 0.627          | 0.635       | 0.509          |
| Plastic Lid ex  | 306          | 372         | 0.591      | 0.575         | 0.599         | 0.515         | 0.588       | 0.578          | 0.587       | 0.458          |
| Empty Paper     | 460          | 569         | 0.651      | 0.671         | 0.67          | 0.557         | 0.645       | 0.666          | 0.655       | 0.53           |
| Other Food c    | 133          | 196         | 0.628      | 0.337         | 0.431         | 0.3           | 0.633       | 0.344          | 0.436       | 0.268          |
| Paper Cup       | 215          | 259         | 0.707      | 0.687         | 0.738         | 0.666         | 0.715       | 0.696          | 0.748       | 0.637          |
| Plastic Drink   | 265          | 355         | 0.835      | 0.758         | 0.859         | 0.747         | 0.84        | 0.763          | 0.862       | 0.737          |
| Batteries       | 1            | 1           | 1          | 0             | 0             | 0             | 1           | 0              | 0           | 0              |
| Plastic Straw   | 148          | 164         | 0.746      | 0.64          | 0.66          | 0.523         | 0.731       | 0.64           | 0.664       | 0.429          |
| Compostable     | 146          | 196         | 0.661      | 0.704         | 0.729         | 0.646         | 0.653       | 0.701          | 0.72        | 0.583          |
| Clear Plastic   | 167          | 189         | 0.752      | 0.755         | 0.788         | 0.717         | 0.752       | 0.757          | 0.792       | 0.711          |
| Sandwich pa     | 126          | 170         | 0.721      | 0.578         | 0.679         | 0.522         | 0.714       | 0.573          | 0.671       | 0.508          |
| Filled Bag      | 203          | 223         | 0.692      | 0.414         | 0.586         | 0.477         | 0.693       | 0.417          | 0.57        | 0.432          |
| Wooden Coff     | 53           | 65          | 0.552      | 0.569         | 0.556         | 0.454         | 0.544       | 0.57           | 0.548       | 0.376          |
| Shelf Stable    | 62           | 135         | 0.705      | 0.745         | 0.78          | 0.653         | 0.699       | 0.74           | 0.773       | 0.639          |
| Flexible cont   | 74           | 91          | 0.779      | 0.385         | 0.465         | 0.391         | 0.734       | 0.364          | 0.444       | 0.365          |
| Magazines N     | 76           | 112         | 0.562      | 0.643         | 0.633         | 0.511         | 0.56        | 0.643          | 0.618       | 0.411          |
| Small Paper     | 22           | 25          | 0.438      | 0.16          | 0.138         | 0.102         | 0.42        | 0.16           | 0.138       | 0.1            |
| Plastic Cutlery | 65           | 78          | 0.487      | 0.256         | 0.329         | 0.241         | 0.483       | 0.256          | 0.282       | 0.16           |
| Receipts and    | 65           | 71          | 0.356      | 0.31          | 0.309         | 0.272         | 0.354       | 0.31           | 0.311       | 0.265          |
| Aluminum Fo     | 45           | 51          | 0.576      | 0.686         | 0.706         | 0.618         | 0.607       | 0.726          | 0.727       | 0.6            |
| Yogurt Tub o    | 47           | 57          | 0.77       | 0.702         | 0.767         | 0.668         | 0.764       | 0.702          | 0.739       | 0.602          |
| Cardboard C     | 65           | 67          | 0.825      | 0.672         | 0.741         | 0.652         | 0.823       | 0.672          | 0.737       | 0.644          |
| Tea Bags        | 32           | 33          | 0.608      | 0.283         | 0.354         | 0.247         | 0.677       | 0.318          | 0.383       | 0.237          |
| Colored Men     | 44           | 54          | 0.417      | 0.389         | 0.336         | 0.302         | 0.41        | 0.389          | 0.336       | 0.291          |
| Clean Paper     | 41           | 56          | 0.763      | 0.46          | 0.588         | 0.5           | 0.734       | 0.444          | 0.56        | 0.463          |
| Glass Bottles   | 70           | 117         | 0.758      | 0.658         | 0.736         | 0.588         | 0.756       | 0.658          | 0.729       | 0.574          |
| Metallic Bottl  | 49           | 59          | 0.641      | 0.454         | 0.515         | 0.361         | 0.617       | 0.438          | 0.501       | 0.322          |
| Compostable     | 34           | 37          | 0.718      | 0.552         | 0.607         | 0.553         | 0.719       | 0.555          | 0.6         | 0.52           |
| Wrapping Pa     | 95           | 114         | 0.67       | 0.351         | 0.507         | 0.372         | 0.666       | 0.351          | 0.481       | 0.327          |
| Compostable     | 37           | 39          | 0.636      | 0.615         | 0.653         | 0.549         | 0.63        | 0.615          | 0.628       | 0.496          |
| Coffee Groun    | 71           | 90          | 0.83       | 0.844         | 0.902         | 0.781         | 0.813       | 0.833          | 0.896       | 0.758          |
| Plastic Milk J  | 32           | 33          | 0.791      | 0.606         | 0.717         | 0.682         | 0.79        | 0.606          | 0.717       | 0.686          |
| Latex Gloves    | 19           | 29          | 0.762      | 0.759         | 0.783         | 0.641         | 0.757       | 0.759          | 0.773       | 0.664          |
| Shredded Pa     | 63           | 78          | 0.833      | 0.744         | 0.797         | 0.707         | 0.833       | 0.744          | 0.796       | 0.716          |
| Refrigerated    | 17           | 21          | 0.596      | 0.571         | 0.559         | 0.474         | 0.644       | 0.619          | 0.567       | 0.481          |
| Liquids         | 8            | 8           | 0.791      | 0.478         | 0.472         | 0.313         | 0.585       | 0.357          | 0.285       | 0.123          |
| Overall:        | Box(Precisio | Box(Recall) | Box(mAP50) | Box(mAP50-95) | Mask(Precisio | Mask(Recall)  | Mask(mAP50) | Mask(mAP50-95) |             |                |
|                 | 0.672        | 0.553       | 0.598      | 0.495         | 0.664         | 0.551         | 0.587       | 0.462          |             |                |

Detailed Comparison of Model Performance

# Solution Approach: AI Image Generation

We implemented a ready-to-use pipeline to fine-tune stable diffusion on classes in the Spare-it dataset to support future work

```
dataset_root = '#path to the crop  
#Dictionary of classes and correspondences  
classes = {  
    "55": "a used teabag",  
    "6": "used latex gloves",  
    "77": "shredded paper",  
    #Add more classes if desired  
}
```

Sample outputs of the  
fine-tuned model



Sample outputs of the  
base model  
(stable diffusion 2.1)





# **Future Improvement**

**S**

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# Future Improvements

## Deploy YOLOv8 for Real-Time Monitoring

We currently have our best YOLOv8 model deployed to Hugging Face, but the ultimate goal of this effort is to support real-time waste analytics. This would likely involve integrating YOLOv8 with a real-time detection system, such as a camera periodically capturing images of a trash can.

## Integrate Synthetic Data

We have implemented a ready-to-use pipeline for fine tuning stable diffusion to generate synthetic data for the Spare-it dataset. Future improvements could be aimed at integrating the synthetic data into the training pipeline, potentially through the copy-paste augmentation.

## More Training Experiments

We did not exhaust all of the possible configurations for training using the tools we built. For example, future work could try and use the copy-paste augmentation only on specific classes in the dataset or implement hyperparameter optimization.



05

# Demo

Link: [Demo](#)

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**Thank You!**